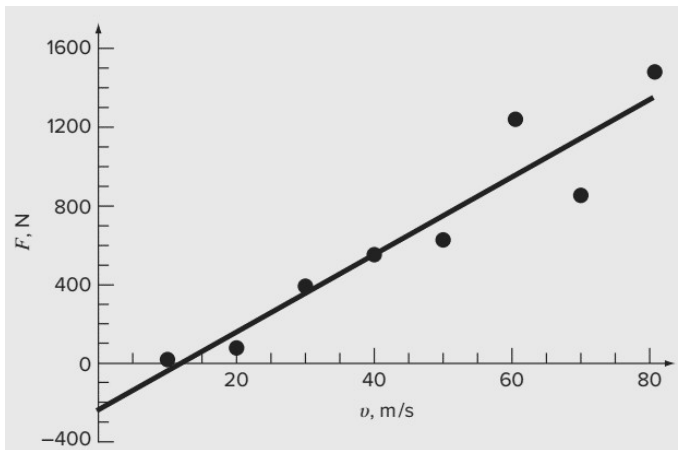
LAB 6: Polynomial Approximation

Objective: To obtain the linear and polynomial approximation (curve fitting) for a given data set.

Theory: Given a set of points (x_i, y_i) for i=0,1,2,…, n, where the x_i are distinct values of the independent variable and the y_i are corresponding values of some function f, Either Approximate the value of y at some value of x not listed among the x_i or Determine a function g that in some sense approximate the data.



The mathematical expression for the straight line is

$$y = a_0 + a_1x + e$$

where a_0 and a_1 are coefficients representing the intercept and the slope, respectively, and e is the error, or residual, between the model and the observations, which can be represented by rearranging Eq. as

$$e = y - a_0 - a_1x$$

Thus, the residual is the discrepancy between the true value of y and the approximate value, $a_0 + a_1x$, predicted by the linear equation. The approach is to minimize the sum of the squares of the residuals as,

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - a_0 - a_1x_i)^2$$

This criterion, which is called *least square method*, has a number of advantages, including that it yields a unique line for a given set of data. To determine values for a_0 and a_1, Eq. is differentiated with respect to each unknown coefficient:

$$\frac{\partial S_r}{\partial a_0} = -2\sum(y_i - a_0 - a_1x_i)$$

$$\frac{\partial S_r}{\partial a_1} = -2\sum[(y_i - a_0 - a_1x_i)x_i]$$

all summations are from i = 1 to n. Setting these derivatives equal to zero will result in a minimum S_r. If this is done, the equations can be expressed as,

$$0 = \sum y_i - \sum a_0 - \sum a_1 x_i$$

$$0 = \sum x_i y_i - \sum a_0 x_i - \sum a_1 x_i^2$$

Upon rearranging,

$$n \quad a_0 + \left(\sum x_i\right) a_1 = \sum y_i$$

$$\left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1 = \sum x_i y_i$$

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - \left(\sum x_i\right)^2}$$

$a_0 = \bar{y} - a_1 \bar{x}$ where $\bar{y}$ and $\bar{x}$ are the means of y and x, respectively.

Problem - Fit a straight line to the values in Table given below,

x_i	y_i
10	25
20	70
30	380
40	550
50	610
60	1,220
70	830
80	1,450

Solution – Following values needs to be calculated-

i	x_i	y_i	x_i^2	$x_i y_i$
1	10	25	100	250
2	20	70	400	1,400
3	30	380	900	11,400
4	40	550	1,600	22,000
5	50	610	2,500	30,500
6	60	1,220	3,600	73,200
7	70	830	4,900	58,100
8	80	1,450	6,400	116,000
$\sum$	360	5,135	20,400	312,850

The means can be computed as

$$\bar{x} = \frac{360}{8} = 45 \qquad \bar{y} = \frac{5{,}135}{8} = 641.875$$

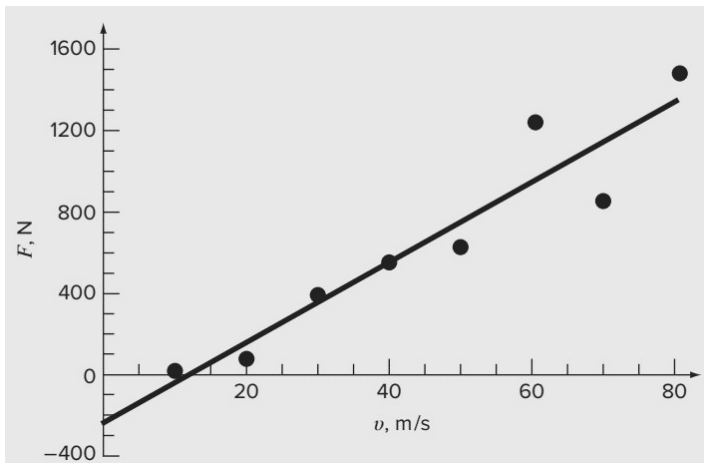
The slope and the intercept can then be calculated with Eqs. (14.15) and (14.16) as

$$a_1 = \frac{8(312{,}850) - 360(5{,}135)}{8(20{,}400) - (360)^2} = 19.47024$$

$$a_0 = 641.875 - 19.47024(45) = -234.2857$$

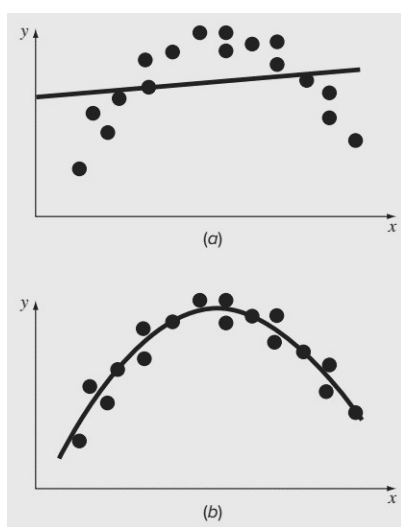
Using force and velocity in place of y and x, the least-squares fit is

$$F = -234.2857 + 19.47024v$$



In MALTAB, first define X vector and Y vector. Then type cftools (or open curve fitting tool from app menu). Explore the linear/polynomial of various degrees and observe the goodness of fit and r values.

Polynomial Regression-



The least-squares procedure can be readily extended to fit the data to a higher-order polynomial. For example, suppose that we fit a second-order polynomial or quadratic:

$$y = a_0 + a_1 x + a_2 x^2 + e$$

For this case the sum of the squares of the residuals is

$$S_r = \sum_{i=1}^{n} \left(y_i - a_0 - a_1 x_i - a_2 x_i^2\right)^2 \tag{15.2}$$

To generate the least-squares fit, we take the derivative of Eq. (15.2) with respect to each of the unknown coefficients of the polynomial, as in

$$\frac{\partial S_r}{\partial a_0} = -2 \sum \left(y_i - a_0 - a_1 x_i - a_2 x_i^2\right)$$

$$\frac{\partial S_r}{\partial a_1} = -2 \sum x_i \left(y_i - a_0 - a_1 x_i - a_2 x_i^2\right)$$

$$\frac{\partial S_r}{\partial a_2} = -2 \sum x_i^2 \left(y_i - a_0 - a_1 x_i - a_2 x_i^2\right)$$

These equations can be set equal to zero and rearranged to develop the following set of normal equations:

$$(n)a_0 + \left(\sum x_i\right) a_1 + \left(\sum x_i^2\right) a_2 = \sum y_i$$

$$\left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1 + \left(\sum x_i^3\right) a_2 = \sum x_i y_i$$

$$\left(\sum x_i^2\right) a_0 + \left(\sum x_i^3\right) a_1 + \left(\sum x_i^4\right) a_2 = \sum x_i^2 y_i$$

Polynomial Regression

Problem Statement. Fit a second-order polynomial to the data in the first two columns of Table 15.1.

TABLE 15.1 Computations for an error analysis of the quadratic least-squares fit.

x_i	y_i	$(y_i - \bar{y})^2$	$(y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$
0	2.1	544.44	0.14332
1	7.7	314.47	1.00286
2	13.6	140.03	1.08160
3	27.2	3.12	0.80487
4	40.9	239.22	0.61959
5	61.1	1272.11	0.09434
$\sum$	152.6	2513.39	3.74657

Solution. The following can be computed from the data:

$m = 2$ $\quad \sum x_i = 15$ $\quad \sum x_i^4 = 979$

$n = 6$ $\quad \sum y_i = 152.6$ $\quad \sum x_i y_i = 585.6$

$\bar{x} = 2.5$ $\quad \sum x_i^2 = 55$ $\quad \sum x_i^2 y_i = 2488.8$

$\bar{y} = 25.433$ $\quad \sum x_i^3 = 225$

Therefore, the simultaneous linear equations are

$$\begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} 152.6 \\ 585.6 \\ 2488.8 \end{Bmatrix}$$

```
a =
    2.4786
    2.3593
    1.8607
```

Therefore, the least-squares quadratic equation for this case is

$$y = 2.4786 + 2.3593x + 1.8607x^2$$

- The sum of squares of residuals, also called the residual sum of squares:

$$SS_{\text{res}} = \sum_i (y_i - f_i)^2 = \sum_i e_i^2$$

- The total sum of squares (proportional to the variance of the data):

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$$

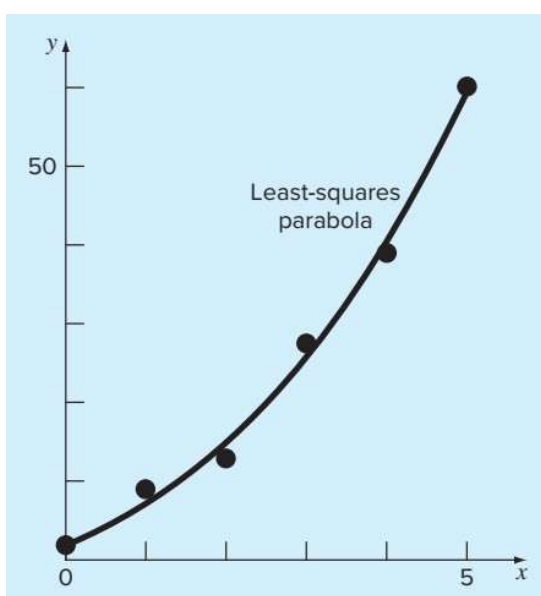
The most general definition of the coefficient of determination is

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

The coefficient of determination is

$$r^2 = \frac{2513.39 - 3.74657}{2513.39} = 0.99851$$

and the correlation coefficient is $r = 0.99925$



MATLAB routine for finding curve fitting (using polyfit function) –

```
clear all; close all; clc;
X = 10:10:80;
Y = [ 25 70 380 550 610 1220 830 1450];
p = polyfit(X,Y,1); % for linear regression
% p = polyfit(X,Y,2) % for qudratic regression
% p = polyfit(X,Y,3) % for cubic regression
scatter(X,Y,'r*');
% make straight line using coefficients found using polyfit function
Y1 = p(2) + p(1)*X;
hold on;
plot(X,Y1,'-b');
```

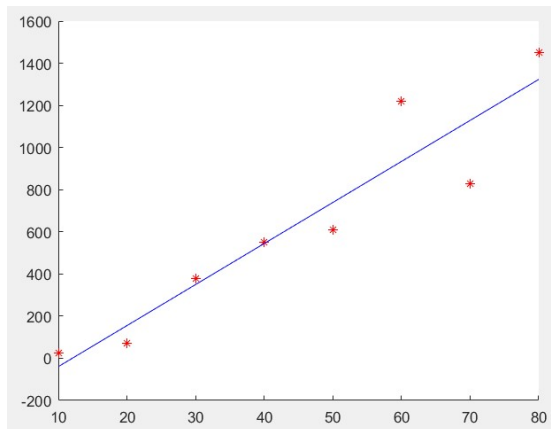


Figure – linear regression of the given data

Alternatively, you can use inbuilt curve fitting tool in MATLAB.

Introducing the Curve Fitting App

You can fit curves and surfaces to data and view plots with the Curve Fitting app.

- Create, plot, and compare multiple fits.
- Use linear or nonlinear regression, interpolation, smoothing, and custom equations.
- View goodness-of-fit statistics, display confidence intervals and residuals, remove outliers, and assess fits with validation data.
- Automatically generate code to fit and plot curves and surfaces, or export fits to the workspace for further analysis.

Fit a Curve

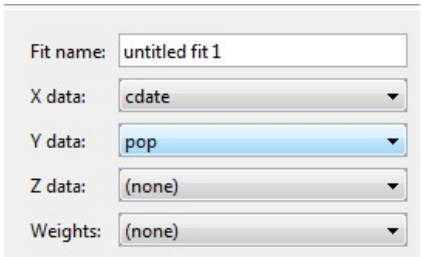
1. Load some example data at the MATLAB® command line:

   load census.mat

2. Open the Curve Fitting app by entering:

   cftool

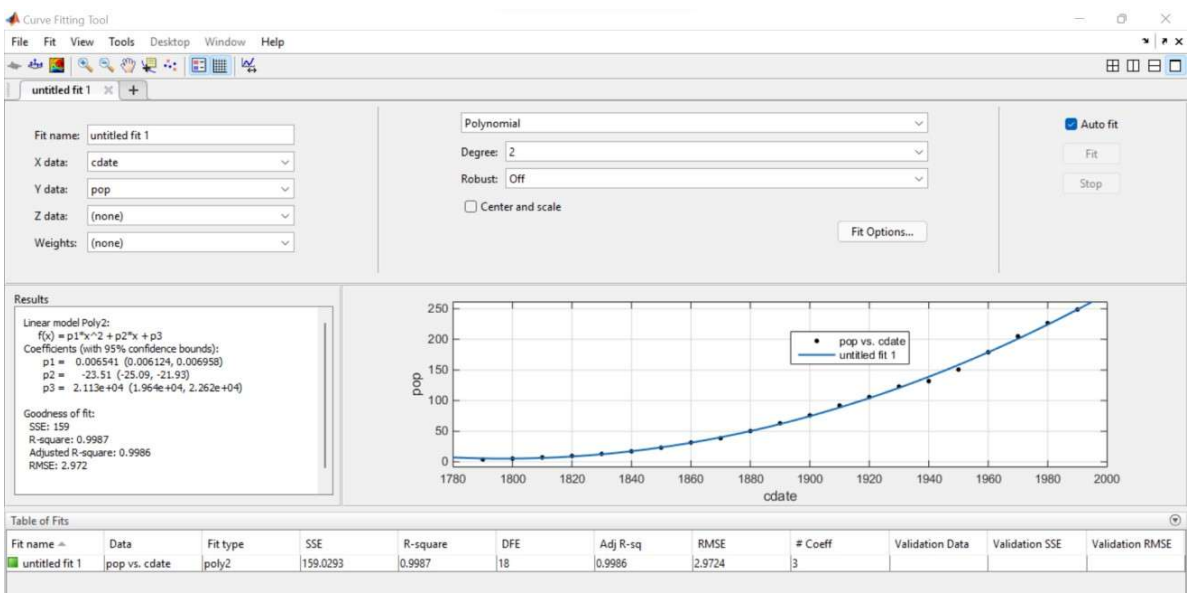
   Alternatively, click **Curve Fitting** on the **Apps** tab.
3. Select **X data** and **Y data**. For details, see Selecting Data to Fit in Curve Fitting App.



The Curve Fitting app creates a default polynomial fit to the data.

4. Try different fit options. For example, change the polynomial **Degree** to 3 to fit a cubic polynomial.

5. Select a different model type from the fit category list, e.g., **Smoothing Spline**. For information about models you can fit, see Model Types for Curves and Surfaces.



Assignment -

One numerical as suggested by Faculty.

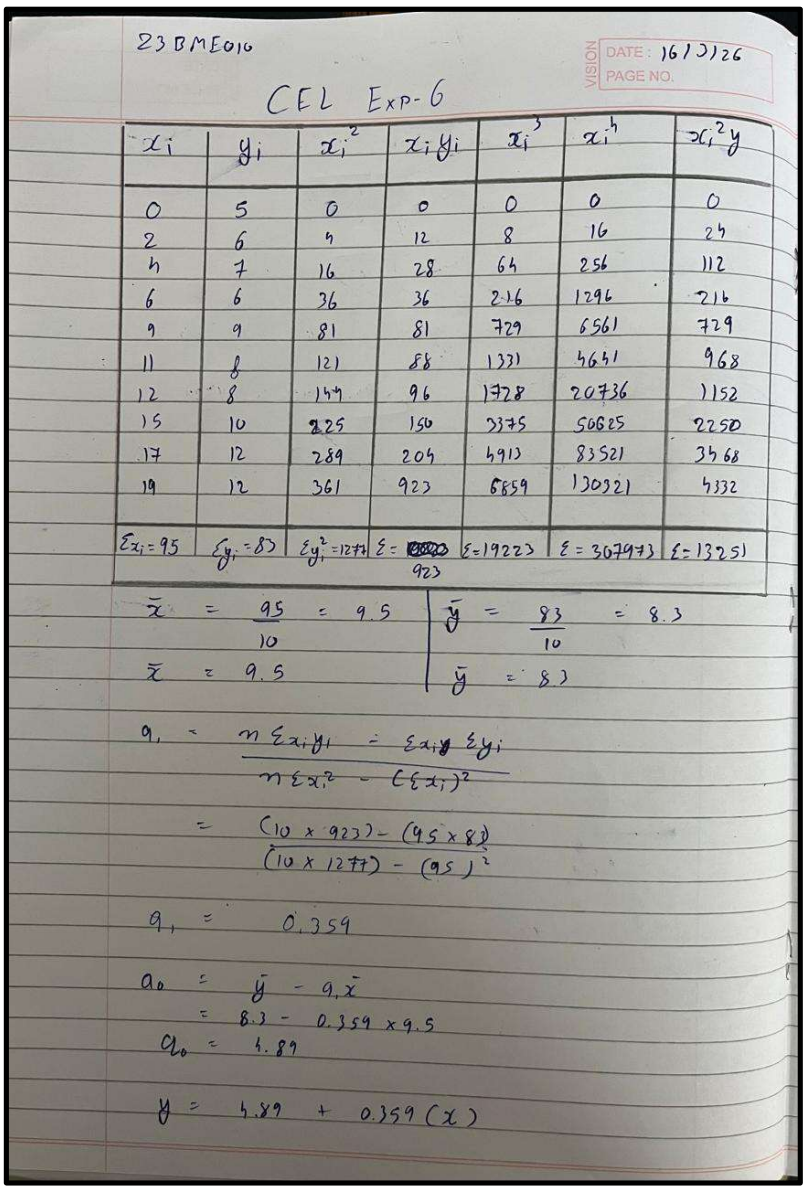
Sample numerical.

Q. Suppose we have xdata and ydata points observed in any experiment as follows –

x	0	2	4	6	9	11	12	15	17	19
y	5	6	7	6	9	8	8	10	12	12

Using hand calculations, show the linear regression for the above data also using curve fitting toolbox in MATLAB, evaluate the goodness of fits with linear and polynomial fit with 2nd degree.

A. Hand calculations:



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x_i	y_i	x_i^2	$x_i y_i$	x_i^3	x_i^4	$x_i^2 y$
0	5	0	0	0	0	0
2	6	4	12	8	16	24
4	7	16	28	64	256	112
6	6	36	36	216	1296	216
9	9	81	81	729	6561	729
11	8	121	88	1331	4641	968
12	8	144	96	1728	20736	1152
15	10	225	150	3375	50625	2250
17	12	289	204	4913	83521	3468
19	12	361	923	6859	130321	4332
$\Sigma x_i = 95$	$\Sigma y_i = 83$	$\Sigma y_i^2 = 1277$	$\Sigma = 923$	$\Sigma = 19223$	$\Sigma = 307973$	$\Sigma = 13251$

$\bar{x} = \frac{95}{10} = 9.5$ | $\bar{y} = \frac{83}{10} = 8.3$

$\bar{x} = 9.5$ | $\bar{y} = 8.3$

$a_1 = \frac{n \Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n \Sigma x_i^2 - (\Sigma x_i)^2}$

$= \frac{(10 \times 923) - (95 \times 83)}{(10 \times 1277) - (95)^2}$

$a_1 = 0.359$

$a_0 = \bar{y} - a_1 \bar{x}$

$= 8.3 - 0.359 \times 9.5$

$a_0 = 4.89$

$y = 4.89 + 0.359(x)$

Figure 6.1 Hand calculation page 1.

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$\rightarrow \Sigma y = n a_0 + a_1 \Sigma x + a_2 \Sigma x^2$ (Eqⁿ - 1)

$\rightarrow \Sigma x_i y_i = a_0 \Sigma x + a_1 \Sigma x^2 + a_2 \Sigma x^3$ (Eqⁿ - 2)

$\rightarrow \Sigma x_i^2 y_i = a_0 \Sigma x_i^2 + a_1 \Sigma x_i^3 + a_2 \Sigma x^4$ (Eqⁿ - 3)

Now, Eqⁿ 1 $\rightarrow 83 = a_0(10) + a_1(95) + a_2(1277)$

Eqⁿ 2 $\rightarrow 923 = a_0(95) + a_1(1277) + a_2(19223)$

Eqⁿ 3 $\rightarrow 13251 = a_0(1277) + a_1(19223) + a_2(307973)$

Solving Eqⁿ ①, ② & ③ we get,

$a_0 = 5.409$

$a_1 = 0.1718$

$a_2 = 0.00989$

$y = 5.409 + 0.1718x + 0.00989x^2$

$(y_i - \bar{y})^2$	$\hat{y}$	$(y_i - \bar{y})^2$
10.89	5.409	0.163
5.29	5.787	0.045
1.69	6.249	0.564
5.29	6.789	0.622
0.49	7.753	1.555
0.09	8.473	0.224
0.09	8.828	0.686
2.89	9.967	0.001
13.69	10.711	1.496
13.69	11.696	0.109
$\Sigma = 54.10$	$\Sigma =$	$\Sigma = 5.465$

Figure 6.2 Hand calculation page 2.

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$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

$$= 1 - \frac{5.465}{54.10}$$

$R^2 = 0.899 \rightarrow R = 0.9481$

$r = 94.81\%$

Figure 6.3 Hand calculation page 3.

Use of MATLAB for curve fitting:

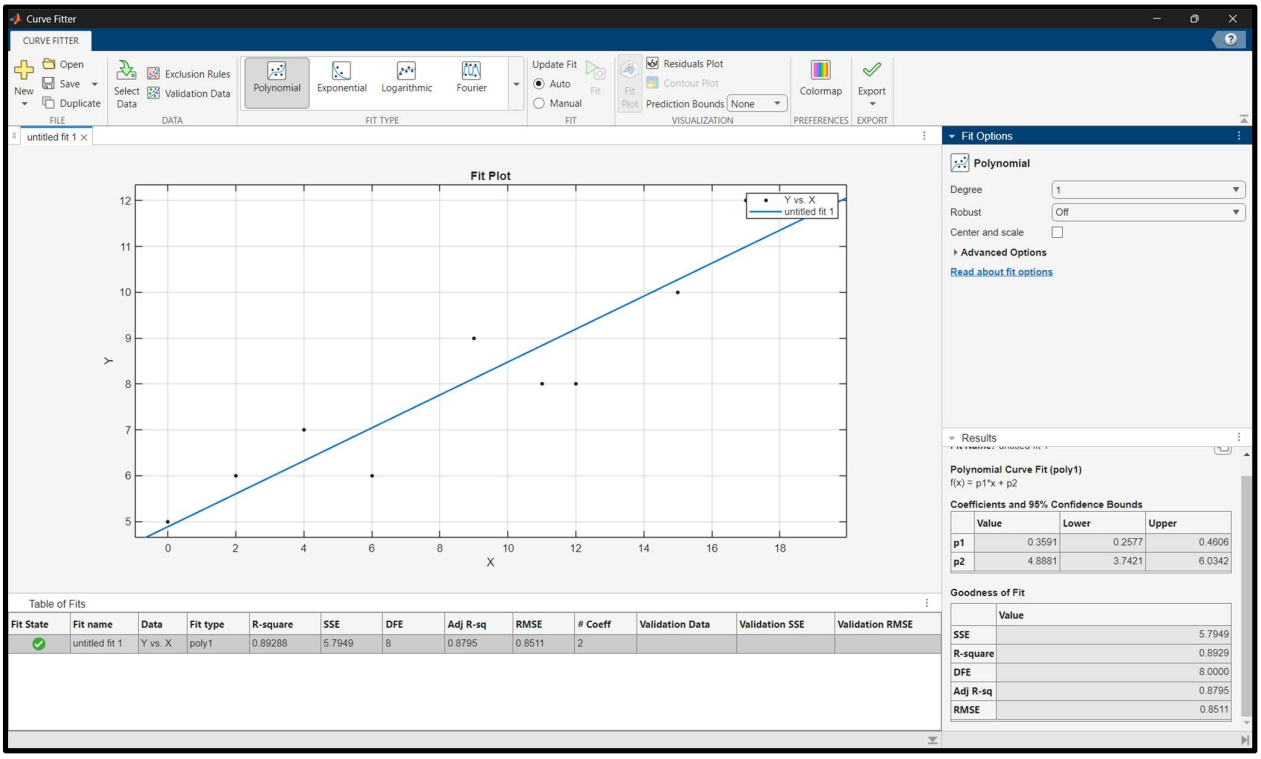

Figure 6.4 Linear Fit Curve

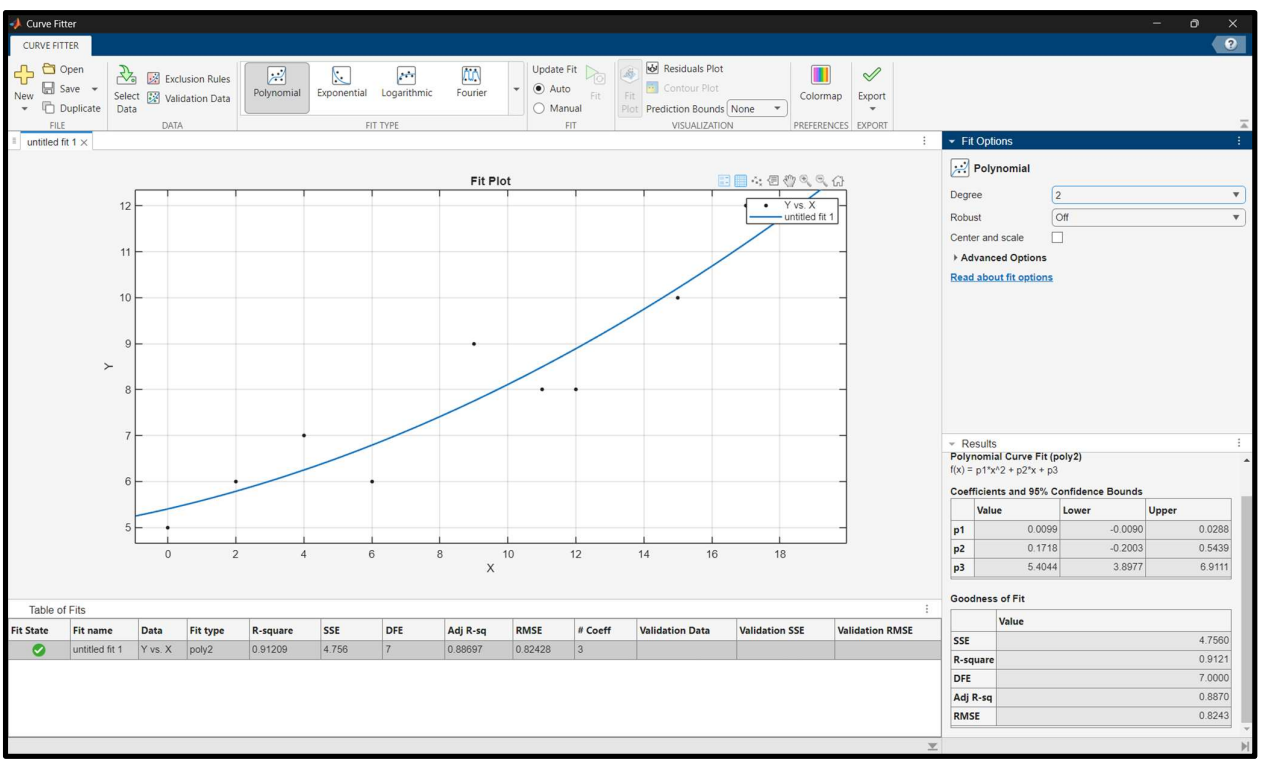

Figure 6.5 Polynomial Fit with 2nd degree

Conclusion:

The given experimental data was analysed using the Curve Fitting Toolbox in MATLAB to obtain both a linear regression model (1st-degree polynomial) and a second-degree polynomial model. For the linear fit, the obtained regression equation was:

$$y = 0.3591x + 4.8881$$

- The goodness-of-fit parameters obtained from MATLAB were:
  - $R^2 = 0.8929$
  - SSE = 5.7949
  - RMSE = 0.8511

This indicates that approximately 89.29% of the variation in the data is explained by the linear model.

For the second-degree polynomial fit, the regression equation obtained was:

$$y = 0.00989x^2 + 0.1718x + 5.404$$

- The goodness-of-fit parameters were:
  - $R^2 = 0.9121$
  - SSE = 4.7560
  - RMSE = 0.8243

The higher R^2 value and lower SSE and RMS (Root mean square) for the quadratic model indicate that the second-degree polynomial provides a better fit to the data compared to the linear model.

Therefore, while the linear regression provides a reasonable approximation of the trend, the quadratic polynomial model more accurately represents the relationship between $x$and $y$for the given dataset.